

AMENDMENT NO. 2 AUGUST 2022
TO
IS 7328 : 2020 SPECIFICATION FOR POLYETHYLENE MATERIAL FOR
MOULDING AND EXTRUSION
(Second Revision)

(Foreword, Para 10) — Insert the following after para 10:

‘Polyethylene manufacturing is a continuous process of polymerization and capable to produce numerous types of grades based upon melt flow index and density which are used by final converters for manufacturing extruded and moulded products (flexible films, raffia tapes, mono filaments, pipes, blow, injection, roto moulded products and various other products).

Since polymerization is a continuous process, during transition from one grade to another, there will be virgin polymer grades produced with broad melt flow index and density range with respect to declared specifications by respective resin manufacturers and will be classified as blending resins or grades. These grades are used as blends with other polymer grades by the converters in various applications without affecting performance in the intended end-use application.’

(Page 1, clause 3.1) — Substitute the following for the existing:

‘3.1 Low Density Polyethylene (LDPE) — Polyethylene, manufactured by high pressure processes, which is highly branched (short and long chains) and has a density of 910 kg/m³ to 930 kg/m³ at 23°C (908 kg/m³ to 928 kg/m³ at 27°C).’

(Page 1, clause 3.2) — Substitute the following for the existing:

‘3.2 Linear Low-Density Polyethylene (LLDPE) — Polyethylene, manufactured by low pressure processes, contain significant amounts of short chain branching and small amounts of long chain branching (when compared to low density polyethylene) and normally has a density of 910 kg/m³ to 940 kg/m³ at 23°C (908 kg/m³ to 938 kg/m³ at 27°C).’

(Page 1, clause 3.3) — Substitute the following for the existing:

‘3.3 High Density Polyethylene (HDPE) — Polyethylene, manufactured by low pressure processes, containing very few short chain branching having a density greater than 940 kg/m³ at 23°C (greater than 938 Kg/m³ at 27°C).’

(Page 1, clause 3.4) — Delete.

(Page 1, clause 3.5) — Delete.

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(Page 2, clause **4.2.3**, code Z) — Substitute the last row for code Z as follows:

Code	Applications
Z	Blending/Miscellaneous

(Page 5, clause **4.3**, Typical Example of Designatory Code) — Substitute ‘Designation Code: IS 7328-2B-FBH-BXDA-C05’ for ‘Designation Code: IS 7328-2B-FBH-BDE-C05’.

(Page 5, clause **5.2.1.1**, Para 1) — Substitute the following for the existing para 1:

‘Density of the material shall be designated as per table 1, based on the value as agreed between purchaser and supplier. It shall be determined according to IS 13360 (Part 3/Sec 10)/IS 13360 (Part 3/Sec 11)/ASTM D1505/ASTM D792/ASTM D4703. Sample shall be prepared according to IS 13360 (Part 2/Sec 1)/3.4.2 of ISO 17855-1/5.1.3 of IS 2530.’

(Page 6, clause **5.2.1.1**, Para 4) — Insert the following after para 4:

‘In case of resin or grade used for blending application, designation code of density in data block 4 shall be based on the actual tested value for density of particular batch/lot and ‘intended application or method of processing’ will be codified as ‘Z’ (clause **4.2.3**). The value of density may vary up to $\pm 8 \text{ Kg/m}^3$ from the measured/reported value.’

(Page 6, clause **5.2.1.2**, Para 1) — Substitute the following for the existing para 1:

‘The melt flow rate/melt flow index of the material shall be designated as per Table 3, based on the value as agreed between purchaser and supplier. It shall be determined by the method prescribed in IS 13360 (Part 4/Sec 1)/ASTM D 1238. The value of melt flow rate shall be within ± 20 percent of the specified nominal melt flow rate, if this is 1 or above and shall be within ± 30 percent of the specified nominal melt flow rate, if this is less than 1.’

(Page 6, clause **5.2.1.2**, Para 2) — Insert the following after para 2:

‘In case of resin or grade used for blending application, designation code of melt flow index in data block 4 shall be based on the actual tested value for melt flow index of batch/lot ‘intended application or method of processing’ will be codified as ‘Z’ (clause **4.2.3**). The value of melt flow rate may vary up to ± 40 percent from the measured/reported value.’

(Page 6, clause **5.3**) — Substitute the following for the existing clause:

‘5.3 Additional Requirements for Compounded Materials

5.3.1 Additional properties to be selected as per **5.3.2** for compounded material shall be determined by the characteristics required for processing and product property requirements.

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NOTE — Purchaser should establish his own correlation between the properties of compounded materials and the material property required for achieving the desired results.

5.3.2 Typical Additional Properties

The typical additional properties as given in Table 6 shall be as agreed between purchaser and supplier based upon the requirements of end product properties.’

(Page 6, clause 5.5.6) — Substitute the following for the existing clause:

‘5.4.6 Dispersion of Carbon Black for Black Compound

The dispersion of carbon black shall be considered satisfactory if the material passes the test prescribed in IS 2530/ISO 18553.’

(Page 6, clause 5.5.2) — Insert the following after Clause 5.5.2:

‘5.5.3 The requirements mentioned in 5.5.2 will remain valid as long as the chemical composition and manufacturing process remains the same. In case of any change in chemical composition and/or manufacturing processes, the requirements mentioned in 5.5.2 shall be tested.’

(Page 8, clause 6) — Substitute the following for the existing clause: **‘6 TESTS**

6.1 For product approval in the manufacturing of LDPE/LLDPE/HDPE materials and the compound made thereof, the following shall be tested:

- a) Density (5.2.1.1), and
- b) Melt flow rate or melt flow index (5.2.1.2).

6.2 Product Identification Tests for Compounded Materials

For the compounded materials to establish the end use application properties, the applicable requirement as listed in 5.3.2 shall be tested, by correlating the required properties of the compounded materials.’

(Page 12, Annex B) — Delete.

(Amendment No. 1, Page 2, Annex D) — Substitute ‘(Clause 5.5.2)’ for ‘(Clause 6.4.2)’.

(Amendment No. 1, Page 2, clause D-2) — Insert the following para at the end:

‘The sample preparation shall be as per the intended end-use of material.’

(PCD 12)

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